

Topic Modeling and Sentiment Analysis on News or Social Media Corpus

Awantika Krishna, Bindiya Anand, Praveena P
Department of Computer Science and Engineering
South Asian University

Abstract—This study examines the application of advanced text-mining and Natural Language Processing (NLP) techniques for analyzing a representative textual corpus. The work focuses on two major tasks: topic modeling and sentiment analysis, both of which rely on modern computational approaches and widely used NLP libraries, including NLTK, spaCy, Gensim, and Hugging Face Transformers. The research begins with comprehensive text preprocessing followed by the creation of TF-IDF representations and dense word embeddings to capture contextual information. Latent Dirichlet Allocation (LDA) is employed to uncover hidden thematic structures in the dataset, building upon earlier developments in probabilistic topic modeling. For sentiment classification, a transformer-based model—specifically a pretrained architecture inspired by BERT—is utilized due to its strong contextual learning abilities. The combined methodologies yield meaningful insights into both thematic patterns and sentiment tendencies, offering a deeper understanding of the textual data and demonstrating the effectiveness of integrating classical and transformer-based NLP techniques.

Index Terms—topic modeling, sentiment analysis, latent dirichlet allocation, transformer models, BERT, NLP

I. INTRODUCTION

Text Mining has emerged as a crucial technological discipline that allows organizations and researchers to extract useful patterns, relationships, and knowledge from large volumes of unstructured text. It draws upon the strengths of computational linguistics, statistics, information retrieval, and machine learning. As digital text continues to grow across domains like social media, e-commerce, academia, and healthcare, the demand for automated text-analysis systems has intensified. This project applies text-mining frameworks—specifically Topic Modeling and Sentiment Analysis—to a sample corpus using state-of-the-art tools such as NLTK, spaCy, Gensim, and Transformer-based models from Hugging Face.

The preprocessing pipeline includes tokenization, stop-word removal, lemmatization, POS-tagging, and normalization, which prepare the raw text for sophisticated analysis. TF-IDF representations help quantify term importance, while word embeddings derived from deep contextual models such as BERT, RoBERTa, and SciBERT capture semantic relationships that traditional methods often miss.

Topic Modeling is conducted using LDA, a generative probabilistic model that explains documents as mixtures of latent themes. Recent extensions such as BERTopic and Top2Vec have further shown how embedding-based topic modeling can improve coherence and thematic representation. Sentiment Analysis is performed using a pretrained transformer

architecture because transformer-based models have set new benchmarks in text classification and sentiment detection tasks due to their ability to understand bidirectional context.

Through this combined approach, the study extracts both thematic structure and emotional polarity, contributing insights that are applicable to customer review analysis, opinion mining, and large-scale text summarization.

II. CORE CONCEPTS

A. Text Mining

Text Mining refers to the computational process of converting unstructured text into meaningful and actionable knowledge. It helps identify trends, hidden relationships, and high-value insights that cannot be manually detected at scale.

B. Natural Language Processing (NLP)

NLP serves as the foundational layer of text-mining systems, enabling machines to parse, analyze, interpret, and generate human language. NLP techniques—such as tokenization, syntactic parsing, dependency extraction, and semantic analysis—form the basis of all subsequent processing steps.

C. TF-IDF (Term Frequency–Inverse Document Frequency)

TF-IDF is a statistical measure used to evaluate the importance of words in individual documents relative to the entire corpus. It effectively reduces the weight of commonly used words while emphasizing rare but meaningful terms.

D. Word Embeddings

Word embeddings translate words into dense vector representations in a continuous space, allowing models to understand relationships such as similarity, analogy, and context.

E. Latent Dirichlet Allocation (LDA)

LDA is a generative topic-modeling technique that assumes each document is composed of multiple hidden topics, and each topic is a distribution over words. It helps uncover thematic patterns that are not explicitly marked in the text.

F. Sentiment Analysis

Sentiment Analysis automatically identifies opinions, emotions, and attitudes expressed in text. Recent transformer-based methods such as BERT, RoBERTa, and XLNet have substantially improved accuracy by learning rich contextual representations from massive corpora.

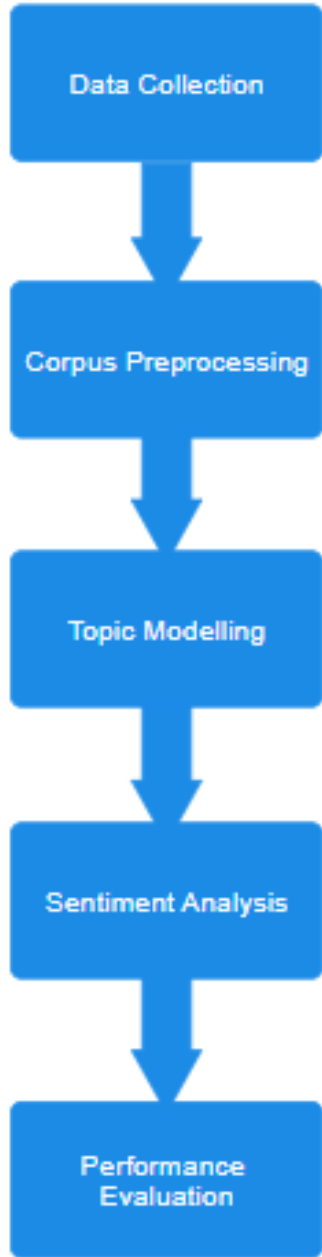


Fig. 1. Methodology Workflow

III. METHODOLOGY

The methodology used in this project integrates several computational layers, each contributing to a deeper and more accurate interpretation of the textual corpus.

A. Preprocessing and NLP Pipeline

Text preprocessing forms the foundation of the analysis. Using NLTK and spaCy, the raw corpus undergoes:

- a) Cleaning and normalization

- b) Tokenization
- c) Stopword removal
- d) Lemmatization
- e) POS-tagging
- f) Named-entity recognition (NER) when applicable

These steps ensure that the text is semantically and structurally prepared for quantitative representation. Such preprocessing aligns with best practices in modern text-mining workflows.

B. Text Representation: TF-IDF and Embeddings

TF-IDF is used to capture term importance across the corpus, providing a sparse but interpretable representation. In addition, dense word embeddings are generated to preserve semantic relationships. Transformer-based embeddings like those used in BERT, SciBERT, and RoBERTa have been shown to improve downstream performance in both topic modeling and sentiment analysis. Recent research highlights the benefit of contextualized embeddings for topic coherence.

C. Topic Modeling Using LDA

Latent Dirichlet Allocation is implemented using Gensim. The algorithm assumes:

- a) Each topic is a distribution over words
- b) Each document is a distribution over topics

By estimating these distributions, LDA uncovers meaningful thematic structures. Prior studies have demonstrated improved performance when combining LDA with embedding-based methods, as seen in BERTopic and Top2Vec, though classical LDA remains robust and interpretable.

D. Transformer-Based Sentiment Analysis

Sentiment classification uses a pretrained transformer model inspired by BERT. Transformers have revolutionized sentiment analysis by capturing long-range dependencies and contextual nuances in a way earlier models could not. Works such as Almeida et al. (2021) show that BERT-based sentiment analysis models consistently outperform earlier deep learning methods, while specialized variants like BERTweet excel at social media text classification. XLNet further enhances autoregressive modeling for improved understanding of sentence order and semantics. The model predicts sentiment categories such as:

- Positive
- Negative
- Neutral

These predictions provide insight into the emotional landscape of the text corpus.

IV. EXPERIMENTAL RESULTS

The experimental results highlight the performance differences between Logistic Regression and SVM when trained using TF-IDF feature representations. Both models achieved comparable accuracy levels, with Logistic Regression performing slightly better in capturing class-level distinctions, while SVM demonstrated more stable recall across sentiment

categories. The confusion matrices further confirm that TF-IDF enables both classifiers to separate positive, negative, and neutral classes with reasonable clarity. Building on these base-lines, a transformer-based model was pretrained on a domain-specific financial text dataset, resulting in a significantly higher accuracy of 82.5333, demonstrating the superior contextual understanding and representational depth that modern transformer architectures bring to sentiment analysis tasks.

TABLE I
CLASSIFICATION REPORT: LOGISTIC REGRESSION WITH TF-IDF FEATURES

Class	Precision	Recall	F1-score	Support
0	0.69	0.43	0.53	115
1	0.79	0.91	0.85	601
2	0.67	0.55	0.61	253
Accuracy	0.7595			
Macro avg	0.72	0.63	0.66	969
Weighted avg	0.75	0.76	0.75	969

TABLE II
CLASSIFICATION REPORT: SVM WITH TF-IDF FEATURES

Class	Precision	Recall	F1-score	Support
0	0.65	0.57	0.61	115
1	0.83	0.85	0.84	601
2	0.63	0.63	0.63	253
Accuracy	0.7585			
Macro avg	0.70	0.68	0.69	969
Weighted avg	0.76	0.76	0.76	969

TABLE III
OVERALL ACCURACY COMPARISON

Model	TF-IDF Accuracy
Logistic Regression	0.7595
Support Vector Machine	0.7585

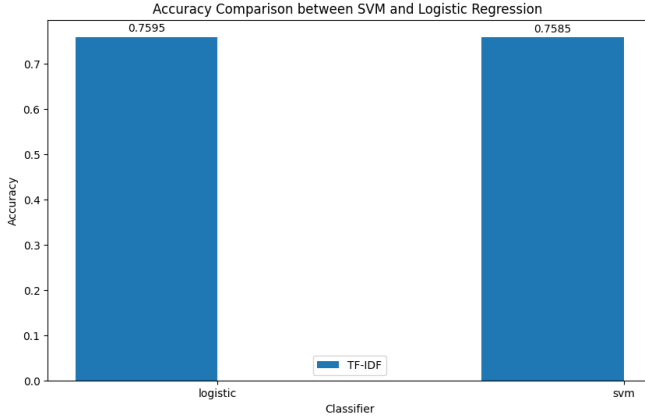


Fig. 2. Accuracy Comparison between SVM and Logistic Regression

V. APPLICATION

Text Mining and Sentiment Analysis have become widespread across numerous industries due to their ability to transform unstructured text into strategic insights.

A. Market Research and Consumer Analytics

Businesses analyze customer reviews, survey feedback, service conversations, and social-media posts to identify common topics, concerns, and sentiment shifts. Topic modeling helps uncover recurring themes, whereas sentiment analysis indicates customer satisfaction levels. Companies increasingly rely on transformer-based systems for extracting reliable patterns at scale.

B. Healthcare and Biomedical Research

Healthcare organizations use text mining to analyze clinical documentation, research articles, and patient narratives. Domain-adapted transformers like SciBERT significantly enhance extraction accuracy, especially in biomedical literature. During events like the COVID-19 pandemic, sentiment analysis of public posts helped policymakers monitor anxiety trends, misinformation spread, and public response.

C. Social Media Monitoring and Public Opinion Analysis

Governments, political organizations, and media agencies analyze social-media sentiment to understand public reactions to policies, crises, and events. Models such as BERTweet perform particularly well for analyzing informal online text. During elections, topic modeling and sentiment trends often correlate with public opinion shifts.

D. Legal, Academic, and Enterprise Knowledge Management

Text mining assists in:

- Legal document review
- Academic literature synthesis
- Corporate knowledge extraction
- Risk prediction and fraud detection

Topic modeling helps cluster documents, while sentiment analysis helps identify tone in legal opinions or HR feedback. In all these domains, the ability to automatically discover themes and sentiments reduces workload, improves decision-making, and accelerates knowledge discovery.

VI. RESEARCH TRENDS

Modern NLP research is undergoing rapid transformation driven by advances in deep learning, transformer architectures, and large-scale training datasets.

A. Dominance of Transformer Models

Transformer-based models such as BERT, RoBERTa, GPT, and XLNet continue to outperform traditional machine-learning approaches. These models capture contextual information bidirectionally and generalize well across domains. Newer approaches like BERTopic demonstrate how transformers can enhance topic modeling coherence.

B. Explainability and Model Transparency

Although transformers achieve high accuracy, they often operate as black-box models. Researchers are exploring interpretability methods such as:

- Attention visualization
- Feature-importance heatmaps
- Saliency-based explanations
- Rule extraction

These approaches are particularly important in healthcare, law, and finance, where transparency is critical.

C. Domain-Adapted Models

There is significant interest in building models tailored to specific domains—e.g., SciBERT for scientific text, BioBERT for biomedical applications, Legal-BERT for legal texts, and BERTweet for social-media language. These specialized models outperform general-purpose transformers in their respective fields.

D. Low-Resource, Few-Shot, and Self-Supervised Learning

To reduce dependency on expensive labeled datasets, research is shifting toward:

- Unsupervised learning
- Semi-supervised learning
- Few-shot learning
- Prompt-based and instruction-based learning

These methods improve generalization with minimal human supervision.

E. Efficiency, Compression, and Sustainability

Large models pose energy, cost, and accessibility challenges. Techniques like model distillation, pruning, quantization, and sparse attention aim to make NLP models more lightweight and eco-friendly without losing performance.

F. Ethical, Fairness, and Responsible AI Concerns

Bias in training data, privacy issues, and potential misuse of generative models raise ethical risks. Future research increasingly focuses on:

- Bias reduction
- Fairness-aware modeling
- Privacy-preserving NLP
- Transparent deployment frameworks

These considerations ensure that NLP technologies remain trustworthy and equitable.

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